

INMO(24th) – 2009

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABC be a triangle and let P be an interior point such that $\angle BPC = 90^\circ$, $\angle BAP = \angle BCP$. Let M, N be the midpoints of AC and BC respectively. Suppose $BP = 2PM$. Prove that A, P, N are collinear.
 2. Define a sequence $\langle a_n \rangle_{n=1}^{n=\infty}$ as follows: $a_n = \begin{cases} 0 & \text{if the number of positive divisor of } n \text{ is odd} \\ 1 & \text{if the number of positive divisor of } n \text{ is even} \end{cases}$.
(The positive divisors of n include 1 as well as n.). Let $x = 0.a_1a_2a_3\dots$ be the real number whose decimal expansion contains a_n in nth place, $n \geq 1$. Determine the proof whether x is a rational or irrational.
 3. Find all the real numbers x such that $[x^2 + 2x] = [x]^2 + 2[x]$, where $[x]$ denotes the largest integer function.
 4. All the points in the plane are coloured using three colours. Prove that there exists a triangle with vertices having the same colour such that either it is isosceles or its angles are in geometric progression.
 5. Let ABC be an acute angled triangle and let H be its orthocentre. Let $h_{\max}$ denote the largest altitude of the triangle ABC. Prove that $AH + BH + CH \leq 2h_{\max}$.
 6. Let a, b, c be positive real numbers such $a^3 + b^3 = c^3$. Prove that $a^2 + b^2 - c^2 > 6(c - a)(c - b)$.

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